Neural Architecture Search

Sean Curow, Noah Worobec, Ryan Lee

Learning Objectives

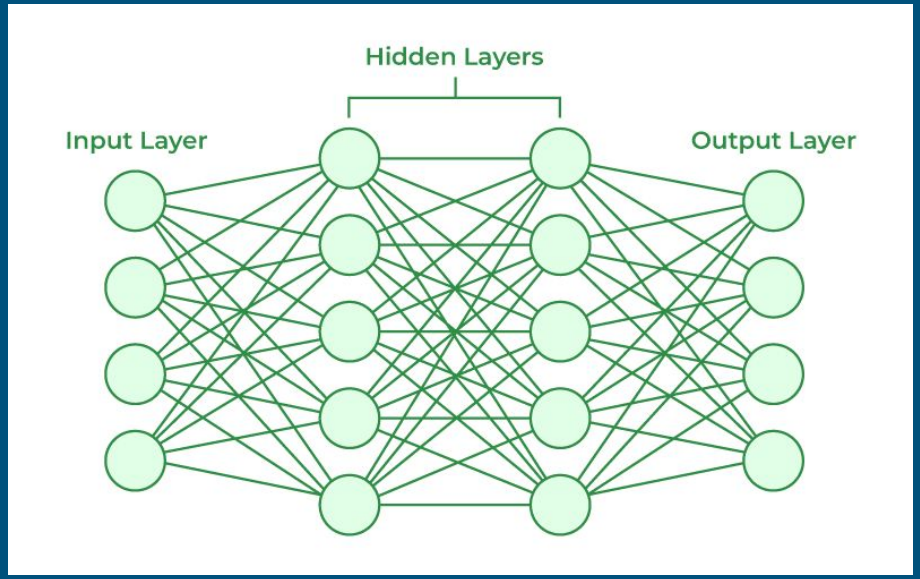
1. Identify the major stages and flow of NAS, explain them in the context of a search problem
2. Understand and Explain the consequences, reasoning, and motivations behind choosing a search space, search strategy, and evaluation strategy for NAS
3. Understand the potential computational cost of NAS
4. Develop the conceptual framework for using NAS to solve a real-world problem

Motivations and Approach

- This topic is vital to modern AutoML, has many uses
  - NLP, computer vision, novel architectures
- This presentation aims to:
  - Break the problem into smaller components
  - Relate to known problem types, simpler structures
  - Provide a complete background of design choices for implementing NAS
  - Briefly analyze the computational cost of NAS
  - Give experimental results and analysis of a minimal implementation of NAS via visualizations and explanations
- This structure was chosen to maximize topic breadth, provide a solid base

What is NAS?

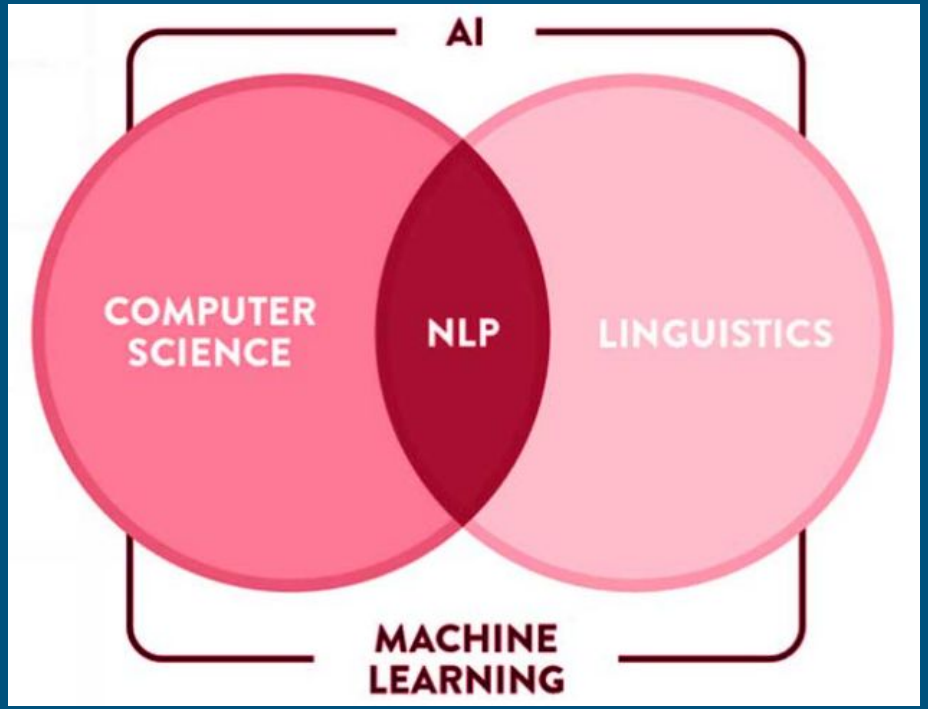
- An AutoML technique, fairly new (~10 years)
- Automating the design of neural networks
  - Required human design before, time consuming!
- Goal is to discover <u>optimal</u> neural networks
- Various strategies create a wide variety of options



https://www.geeksforgeeks.org/deep-learning/artificial-neural-networks-and-its-applications/

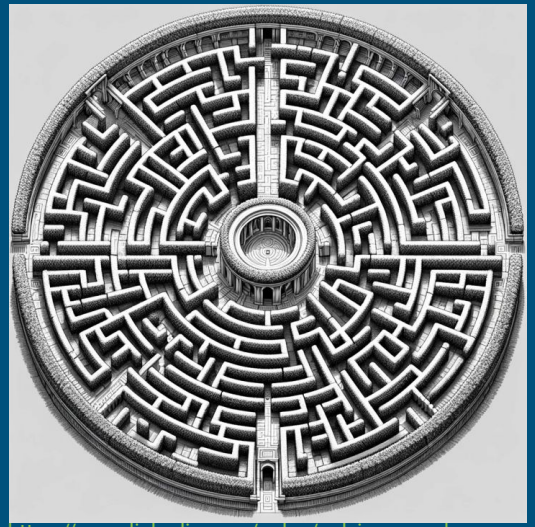
Why NAS?

- From before, human design takes time
  - Automation still takes a while, but frees up resources
- Due to optimization problem nature, can outperform human design!
- Main uses include image recognition, NLP
- Can handle more complex data and solutions than humans (scalability)



https://www.engati.ai/glossary/natural-language-processing

NAS as a Search Problem

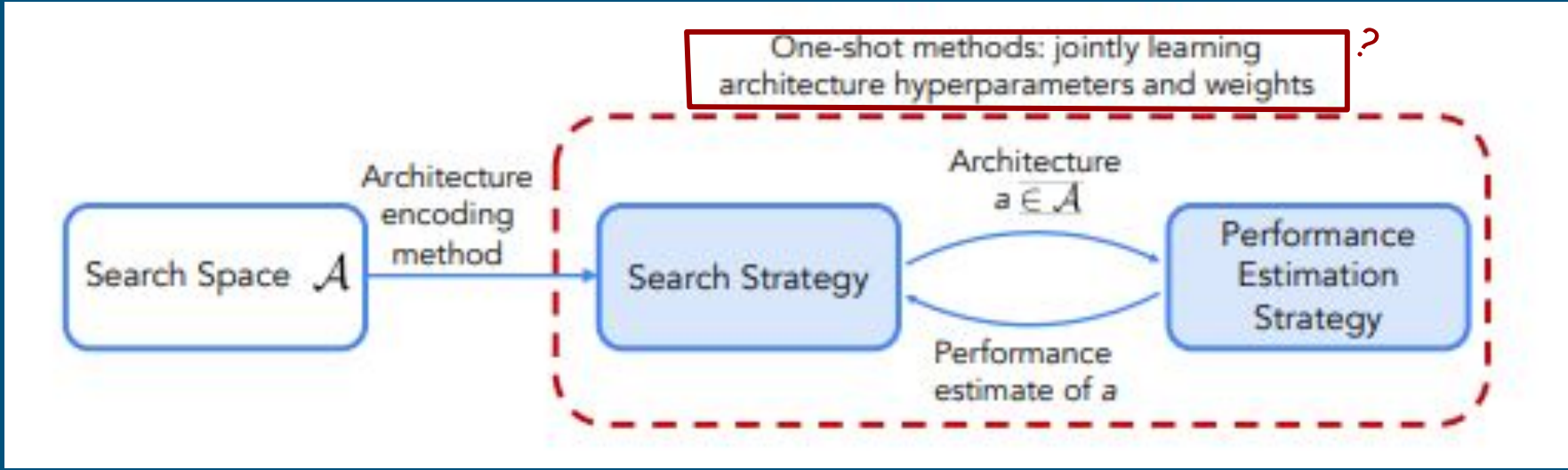


https://www.linkedin.com/pulse/solving-search-problems-artificial-intelligence--pbz1c/

1. Search Space
   a. Set of neural architectures to choose from
2. Search Strategy
   a. Algorithm(s), method(s), to select an architecture
3. Evaluation Strategy
   a. Train architecture, evaluate performance

- Choose (1), loop (2) and (3) until a chosen criteria is met (iterations, performance benchmark, etc.)
- Different search strategies and evaluation strategies can be chosen for different tasks

NAS as a Search Problem



https://arxiv.org/pdf/2301.08727

Search Spaces

- The set of all neural architectures that NAS can choose from when searching
- Size can range from a couple thousand to over 10e20!
- Choosing this can be tricky:
  - Too general, the search time can be massive, time wasted on architectures that would never have worked
  - Too pre-defined, human bias is introduced and the purpose of NAS is slowly lost
- The definition of the search space has massive implications on the results of NAS

Search Strategy

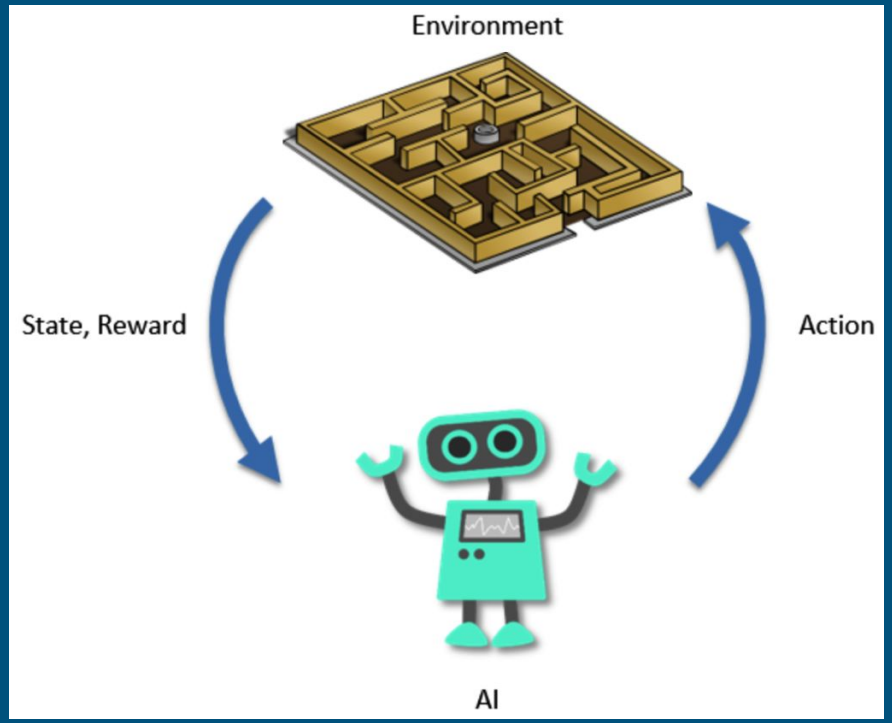
- How do we search the search space?
  - Random
  - Reinforcement learning
  - Bayesian optimization
  - Evolutionary algorithms
  - Each of the above has its uses, though some are clearly favored

https://info.aiim.org/aiim-blog/intelligent-search-strategies

- Random search often a baseline, but it can work well on its own
  - Efficiency decreases as search space size and complexity increases
  - If the entire search space is searchable, guarantees the best architecture IN THAT SPACE

Search Strategy

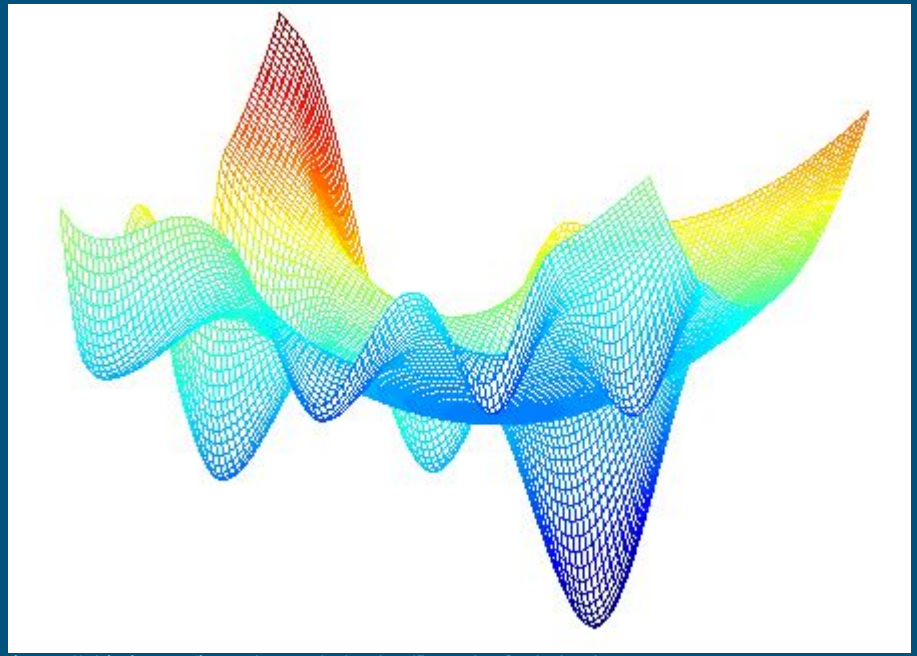
- Reinforcement learning is a formerly popular, but costly solution
  - Hundreds of GPU's, hundreds of hours, high energy and time cost
  - Can work quite well on complex search spaces, efficiency varies with size though
  - Choose architecture, train and evaluate, use evaluation score as "reward" input for the next iteration



https://gordicaleksa.medium.com/how-to-get-started-with-reinforcement-learning-rl-4922fafeaf8c

Search Strategy

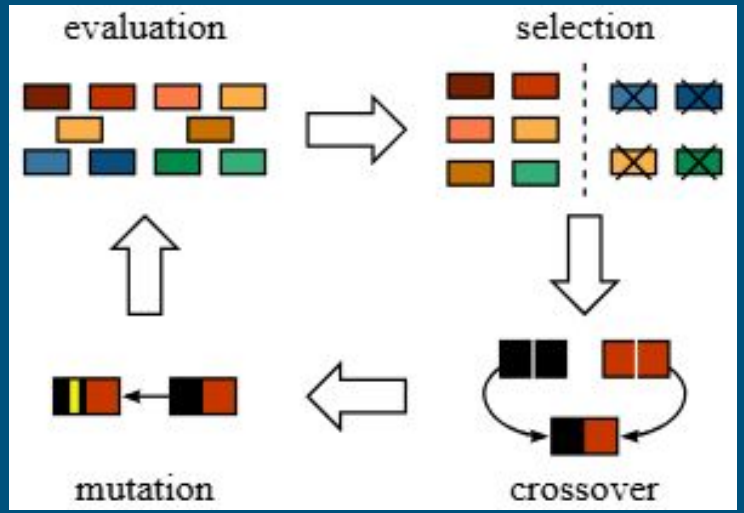
- Bayesian optimization has been a hugely successful approach
  - Simulate the objective (performance) with a probabilistic model, choose the architecture that maximizes the objective, retrain the model with the new architecture
  - Works well with search spaces that are complex to evaluate



https://github.com/bayesian-optimization/BayesianOptimization

Search Strategy

- Evolutionary methods popular for their flexibility and good-enough results
  - Iteratively handle a population of architectures by selecting parents and combining/mutating to produce children
  - Results often outperform random and reinforcement learning most of the time



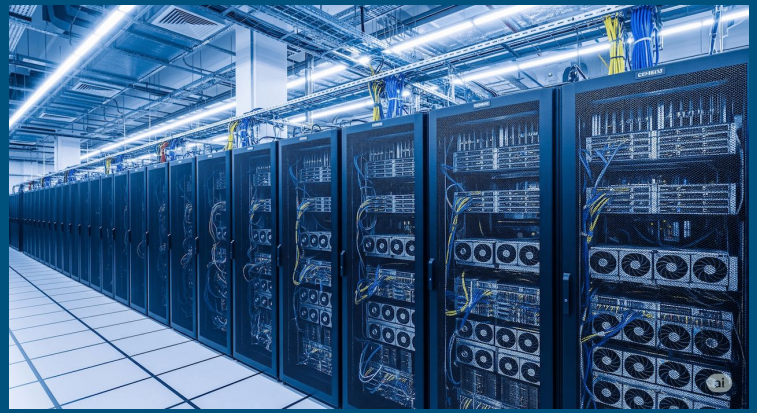
https://www.strong.io/blog/evolutionary-optimization

Evaluation Strategy

- Training a full neural network each iteration will be extremely resource intensive
- How can we mitigate this?
  - Proxy tasks (similar, but simpler)
  - Scaling down (< iterations, weights, training data)
  - Inherit parameters (partially trained, or from previous fully trained)
  - Learning curve extrapolation (if a model is failing, cut)
- Like various search strategies, different evaluation strategies can be better for different tasks
- All of these strategies have risks, like missing a promising model due to abstraction

Computational Cost

- Long story short: Very Expensive!
  - Traditionally, 100s of GPU-days and 100s of GPUs worth of memory
  - Modern methods (one shot) can reduce these figures to 1-5
- One shot creates and trains a “super network” from the search space, instead of individual architectures
  - Architectures then evaluated with weights from their corresponding “sub network”
- NAS is one of the most expensive AutoML techniques



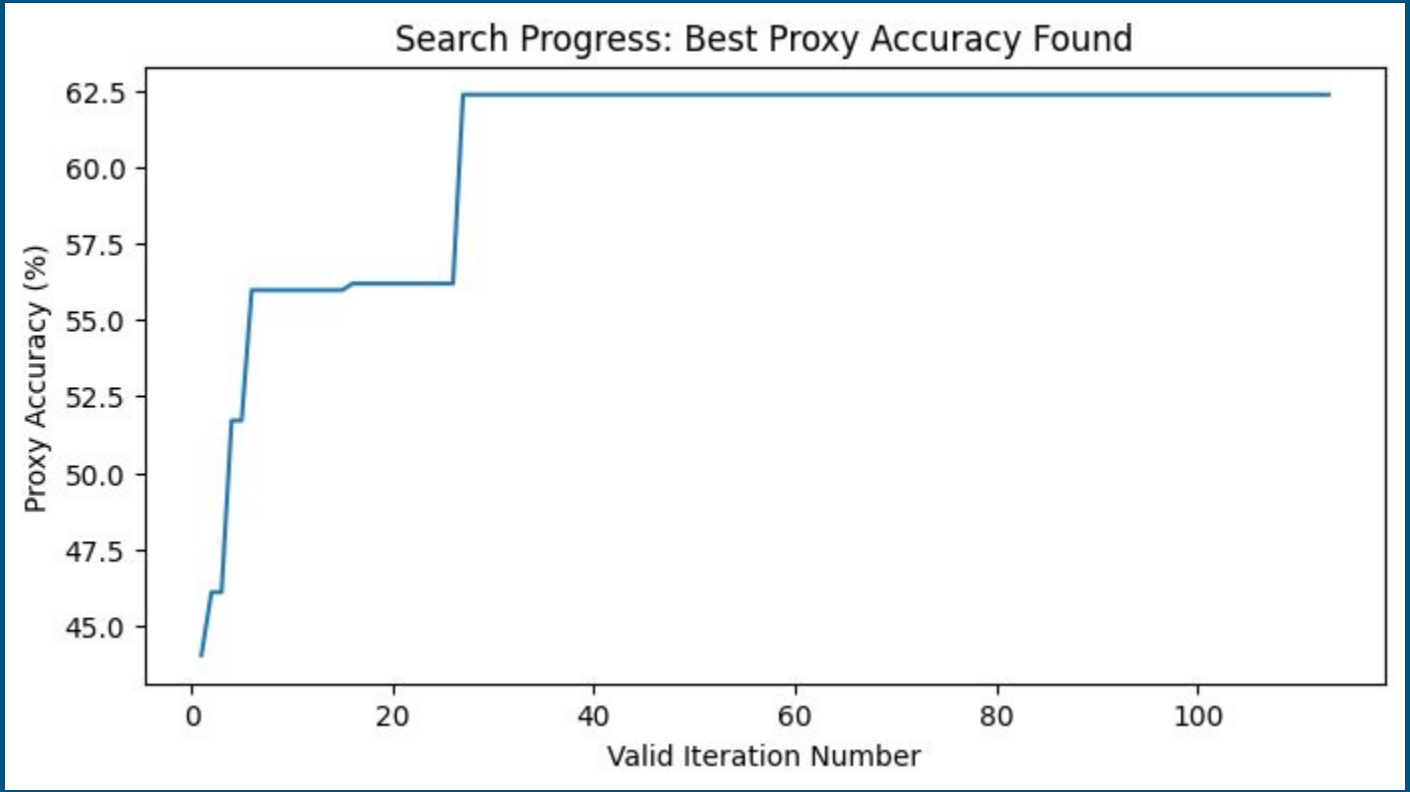
https://appetals.com/blog/the-complete-guide-to-building-gpu-servers/

https://www.amazon.com/Ultimate-Wall-Clock-Battery-Operated/dp/B07HVN4BVQ?th=1

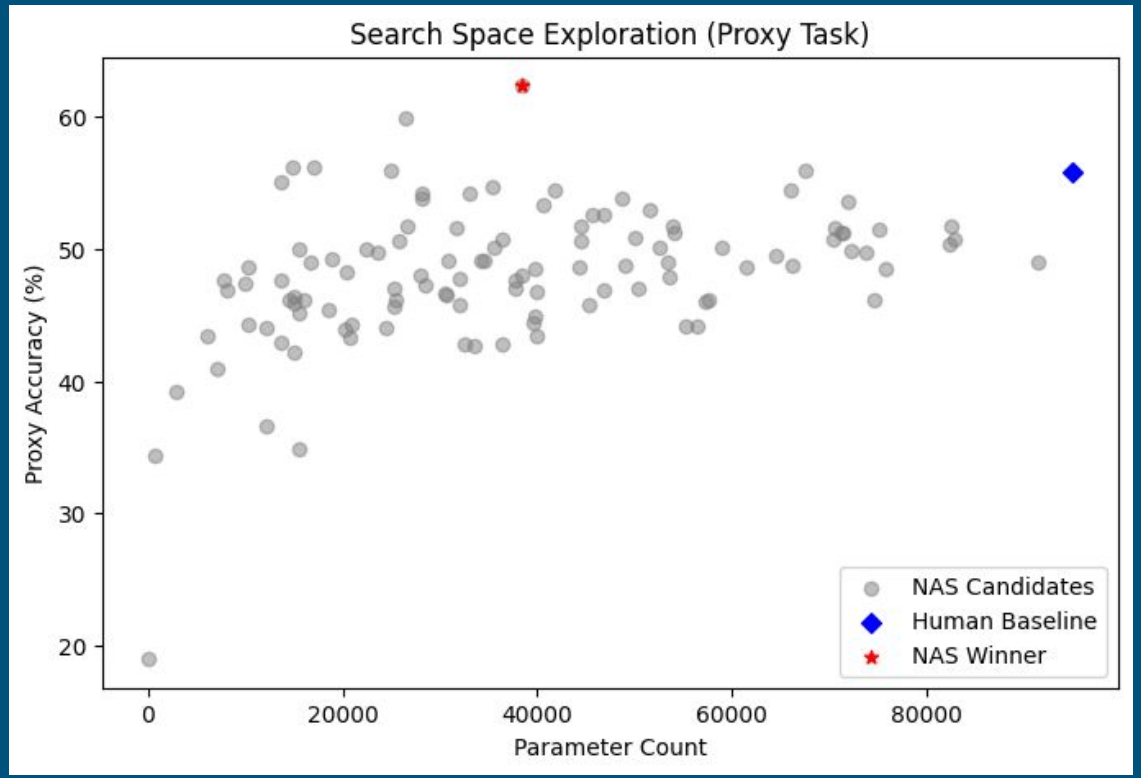
Experimental Results and Analysis

- Using a human-designed baseline model to evaluate the relative performance of a minimal NAS (random search)
- Limit the NAS algorithm to strictly keep its models under the human designed parameter budget
- Search space involves randomly chosen operations compiled into a valid PyTorch model
- Search strategy is random search
- Evaluation strategy uses a proxy function

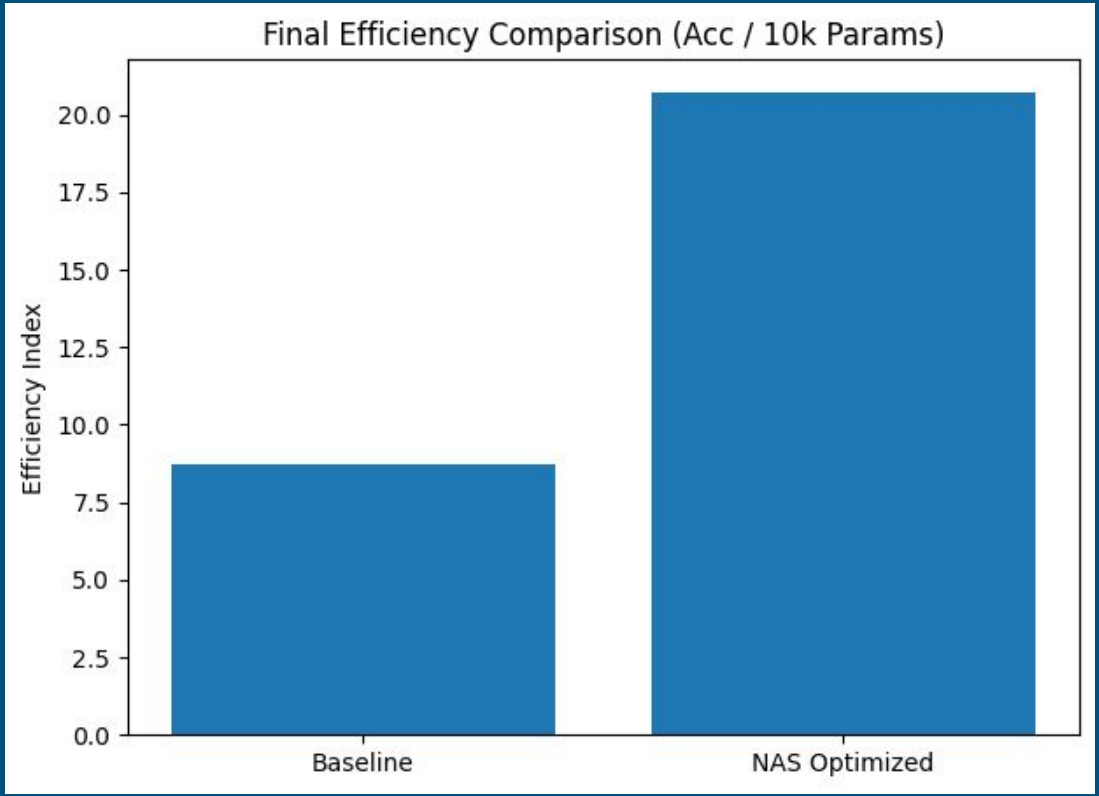
Experimental Results and Analysis



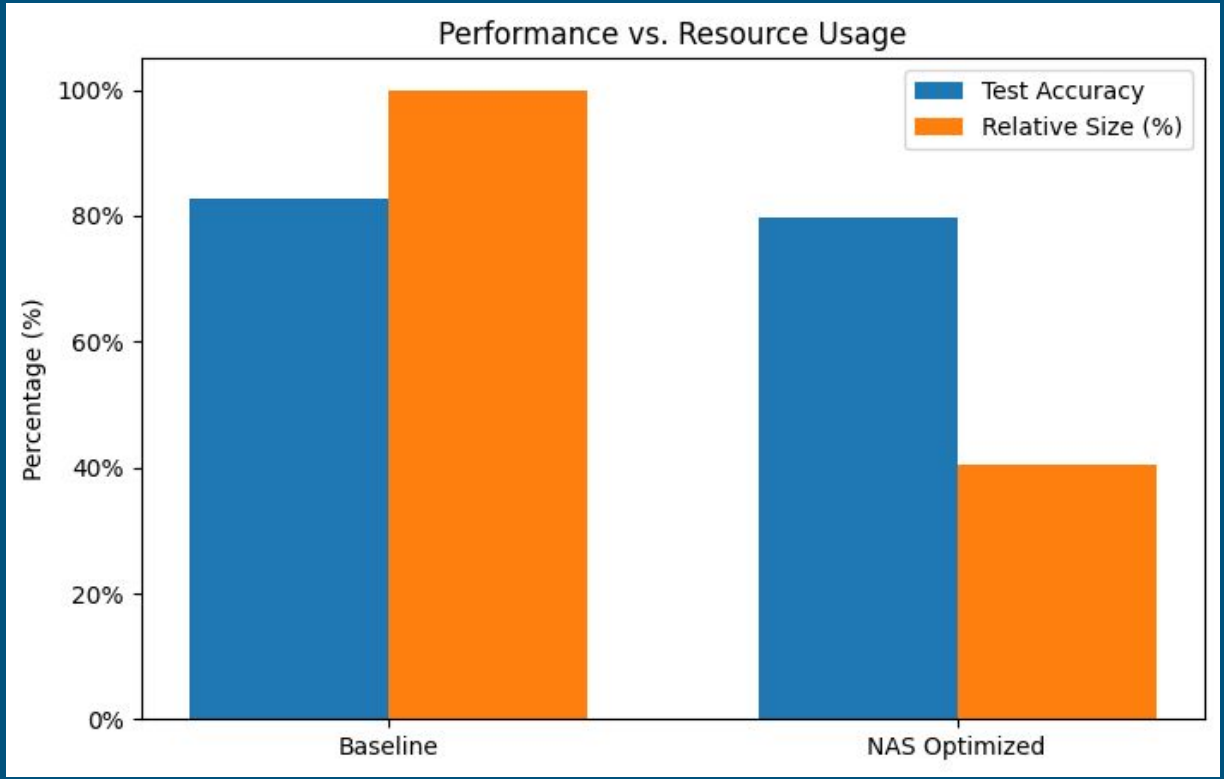
Experimental Results and Analysis



Experimental Results and Analysis



Experimental Results and Analysis



Limitations, Potential Expansions

- NAS is expensive! One-GPU machines are limited to minimal implementations
- With a more sophisticated setup, our experiments could be expanded significantly
- Larger, more complex, and more useful datasets, more useful problems, real world applications
- Try other methods for search, evaluation as well
- Still a lot to do and a lot to learn!

Other Resources

NAS basics: https://www.geeksforgeeks.org/deep-learning/neural-architecture-and-search-methods/

Results from 100 papers on NAS: https://arxiv.org/pdf/2301.08727

Various other papers/resources on the topic: https://www.automl.org/nas-overview/